Dzmitry Lahoda, Software Engineer

I specialize in distributed systems, infrastructure, blockchain, reliable protocols, and high-availability software.

Previously, I worked on enterprise systems, gaming, desktop applications, and device drivers.

I focus on growing code that is scalable, composable, fast, secure, and maintainable — one light day at a time.

Based in Portugal since 2022.

Continuously sharpening my skills in mathematics, compilers, finance, and cryptography.

I have read and applied those old papers where people described what is to be engineer in software.

Profiles

GitHub
LinkedIn
HackerRank Certificate
Skills and tools

Mostly `Rust`. Learning `Lean4`.
Sometimes dig into Nix, Solidity, TS, Python.
Used Go, Object Pascal, Matlab, C#/F#, C within Kubernetes or Windows before.

Work

2024.03.04 - now, N1XYZ, Full time

- Maintained and enhanced the codebase of a **LMAX based central limit order book**
- Designed and implemented **perpetuals funding index**, **reduce orders**, **liquidator mechanics** (with a **sans-io based bot**).
- Reduced bugs and maintenance overhead by introducing several **typestate patterns**
- Executed efficient prop testing strategy (wide coverage with inverses and invariants, and yet fast and low code), bugs found
- Developed a `borsh` schema-based storage diff utility and TS API client generator.
- Designed **manual offline** and **automatic live upgrades**; unified messaging for L1/L2s, storage replication protocol for an **optimistic ZK fraud-proof rollup**, partially implemented
- "Discovered" few Rust patterns which not documented in internet to make code fast, convenient and safe

2022.01-2024.03, Composable Foundation (bankrupt), Full time

Design, development, maintenance and audit of Polkadot parachain, Cosmos CosmWASM VM for Polkadot, cross-chain interoperability, intent-based super app and cross-chain trading/routing infrastructure and middleware (CoW like), IBC cross-chain contracts infrastructure in Ethereum Solidity, lending, AMM, staking/rewards, vote escrowed, governance, indexers, relayers and devtools.

2021.03-2022.03, Web3.0, blockchain, p2p. Software Engineer. Contract works (Fluence Lab, Boosty Labs, MLabsCity, Velora)
Design, coding, audit, research, and deployment of AMM, NFT, crowdfunding, IDO, governance, DAO, identities, indexers, tools, chats for Solana and libp2p.

2021.05-2021.11. The Workplace Metaverse. Software Engineer. Contract works
Coded 2D graphics metaverse engine features.
Prototyped 1000+ user event-driven scalable server for world consisting of dynamic allocation of its parts, with WASM client.

2020.10-2021.03, Elemy, bankrupt. Software Engineer. Contract works
Did a lot of long-term healthcare business flows via CQRS/ES and templated low-level PDF rendering engine for patients to sign.

2020.03 - 2022.06, Metaverse game builder startup (Crey Games), bankrupt. Software Engineer.
Did hybrid global autoscaled cloud native instance allocation, upgrades and matchmaking for game servers;
referral, game event objectives, online presence and other flows into microservice architecture.
Prototyped and scale tested graph database for game economics world (ownership and visibility of various assets and stuff, with roles and users).
Introduced distributed logging and alerting, monitoring dashboards.

2019.11.11–2020.02. eCommerce Point of Sale. Walmart. Software Engineer
Implemented California Consumer Privacy Act conformance (opt-out, access and PII delete flows) with new and within existing microservices and delivered (with ops, monitoring, alerts).
Wrote distributed testing tools, debugged and patched scalability issue in graph database driver.
Made `read-delete-test+read` pattern PII delete playbook with dry run.

2019.08–2019.10. Cloud-native microservice-based call and contact center, Luware. Software Engineer.
Scaled and persisted call handling microservice.
Executed logging storage for 3TB logs storage with 40GB logs per day.

2018.10 - 2019.08.09. Trooper Shooter: Critical Assault FPS (3D mobile shooter). AZUR INTERACTIVE GAMES. Software Engineer
Coded middleware and server parts of game from PoC to Published state.
Did fast real-time networking over UDP, game mechanics middleware via ECS, backend, infra, matchmaking and configuration based meta, actors.
Stress tested my real-time part to prove commodity server thread can game 35+ concurrent users.

2017.10 - 2018.10. Bosch-ETAS Measurement data analyzer. Software Engineer.
Developed parts of app to show and analyze measurement signals from car sensors,
with real-time data synchronization.

Upgraded single-threaded to multithreaded database usage, synchronous blocking buggy code to bug-free reactive, real-time data views.

06.2015 - 08.2017. Thomson Reuters Legal Electronic discovery SaaS. Software Engineer
Legal product aimed to decrease the volume of information within litigation to derive insights and create legal arguments

- Developed asynchronous actor-like data-intensive multi-tenant distributed persisted engine and file system aimed for upload/processing/production/analysis/search/review of millions of documents.
- Developed orchestration with error handling of event-driven services, API, billing, audit, reporting, query provider to NoSQL database, asynchronous data- and work-flows
- Optimized system scaling, data ingestion and access performance on 20TB+ of multi storage indexed data
- Created malicious e-Discovery hardening data set and document relationships visualization

2014.07-2015.01. Accounting and practice management software platform for mid and large law firms. Software Engineer
- Maintained custom client-server application framework, database oriented IDE, ORM.
- Fixed in fault tolerant background task executions and notifications

2011.07-2013.01. Website security/forensics audit for [PKI] usage in Auctions2. Security Researcher
Used reverse engineering to find several holes in site security from client code to DB.

2010.02-2014.06. Office add-ins suite and application framework for financial professionals, Eikon for Office. Desktop Software Engineer
Design and development of linking data between documents, financial charting tool, productivity enhancements for Microsoft Office, framework for an integrated suite of desktop applications used by distributed teams, multi-process data enabled applications connected to several cloud data sources, automatic upgrades, integration of various products into cohesive solution.

2008.10-2010.06, part-time. Music Information Retrieval, visualization and analysis. Researcher
- Prototyped visualization to organize music collection onto 2-dimensional scatter plot via PCA based on audio content processed by FFT and MIRToolbox.
- Researched algorithms which understand music from audio content using Hierarchical Temporal Memory and classify via KNN

2008.07, 2009.07, internship. Time Lapse Microscopy Capture Tool, Institute of Physical Biology
Created Time-lapse microscopy app for camera attached to the microscope for continual shooting, entropy-based quality measurement, tuning camera settings.

2008.08-10, part-time. Social network data analysis and visualization, Itransition. Intern
Coded some server-side part to get data to and from database

2006.08-12, part time. Woodworking machine driver, Stroydetali LLC. Driver/Desktop Software Engineer
Created real-time application for woodworking machine to manage sawing plan, configuration and control of sawing process.
Domain specific configuration file format, system event loop, TUI with edit boxes/menus/labels/help, bit level communication protocol with hardware.

2004.09-2005.03, part-time. Graduates database
Coded graduates app, 2nd place in regional competition of school student's software

Education

2010.04-2014.02, part-time

- Brilliant (many math courses)
- Machine Learning, Model Thinking, Data Science, Functional Programming Principles on Coursera.
- Watched and worked through all lectures in course of Stanford Programming Paradigms (CS 107).

2005.09-2010.06. Belarus State University, Faculty of Radiophysics and Computer Technologies
Bachelor's degree, Major Radiophysics - 01.04.03 GPA: 6/10.

2008, part time. Enterprise/.NET software development courses

1994-2005. Gymnasium №1, Vileyka, Belarus
Physics and mathematics speciality, state olympiad in Physics. GPA: 9/10.